

Computer Networks
CSE1405

Self-Study Assignment

Data Link layer

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Topics covered

- ① Go Back N and Selective Repeat
- ② Sliding Window Protocol Throughput

EXERCISES

Acknowledgments

Question

Erie sends 200-byte frames to Piyush who immediately acknowledges them using 4-byte acknowledgments. The latency of their communication channel is 250ms and the bandwidth is 500 bytes per second in both directions.

How long does it take until Erie receives the acknowledgment from the moment she starts sending a frame (without any errors)?

If we ignore the transmission delay of the acknowledgment, how inaccurate is the computation, i.e., how much does the result differ from the result computed before?

Suggested time for this exercise: 8 minute(s)

Timeout

Question

Aniket and Thorsten use Selective Repeat with a sliding window of $k = 4$ and a timeout of $500ms$. Their channel has a round trip time of $350ms$ and a bandwidth of 500 bytes per second. Frames are always of size 150 bytes and acknowledgments are short so that their transmission delay can be ignored.

Aniket sends frames constantly and one frame is lost but the subsequent frame arrives, as does the corresponding negative acknowledgment. Timers are started immediately once frames have been transmitted.

Does the negative acknowledge arrive at Aniket before the timeout for retransmission of the lost frame?

Suggested time for this exercise: 5 minute(s)

Selective Repeat/Go Back N

Question

Bert sends Ernie 20 100-byte frames using a duplex channel. Acknowledgments are very short, so their transmission delay can be ignored, and a sliding window protocol with $k=5$ is used. The latency is 50ms in both directions and the channel has a bandwidth of 1000 bytes per second in both directions. There is an error when transmitting frame 5 the first time but all other frames and acknowledgements are transmitted correctly.

Assume that when Bert needs to retransmit a frame, he first finishes the transmission of the frame he is currently transmitting if any. He then retransmits before sending other frames in the queue. Which frames are retransmitted when using Go Back N and Selective Repeat, respectively, with a timeout of a) 600ms, b) 210ms, and c) 150ms?

Suggested time for this exercise: 20 minute(s)

► Hint

Sliding Window Frames

Question

Theoden sends 200-byte frames to Faramir using a duplex channel that can handle up 400 bytes per second. The latency of the channel is 230ms from Theoden to Faramir and 270ms from Faramir to Theoden.

They use a sliding window protocol with $k = 4$.

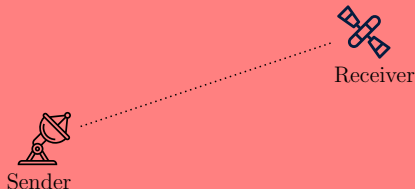
For the first k frames that Theoden sends, does the acknowledgment for the first frame arrive before Theoden is done transmitting the k -th frame?

Assume that there are no errors and Theoden is constantly sending. The transmission delay of the acknowledgment as well as the processing time can be ignored.

Suggested time for this exercise: 5 minute(s)

Sliding Window

Question



Consider an error-free 64-kbps (kilo-bits per second) satellite channel used to send 512-byte data frames in one direction, with very short acknowledgments coming back the other way.

We know that the round trip time from earth to satellite is 540 ms.

Find the throughput for window sizes 1, 7, 15, and 127.

(Assume the processing time at the receiver side and the transmission delay of the acknowledgment can be ignored.)

Suggested time for this exercise: 25 minute(s)

Congratulations!!

By completing this tutorial, you mastered:

- analyzing the difference between different sliding window protocols
- calculating throughput in sliding window protocols
- turning bytes into frames

Hints

Hint: Selective Repeat/Go Back N

Hint

A negative acknowledgement in Selective Repeat does not count as an acknowledgement of the subsequent frame, i.e., a negative acknowledgement for frame 5 does not acknowledge frame 6.

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Hint: Sliding Window

Hint

If the acknowledgment for a frame arrives before the following $k - 1$ frames are transmitted, the sender does not have to stop sending. Otherwise, they have to stop until the acknowledgment arrives.

The computation differs depending on which of the two cases you are in. So, first establish which case it is and then compute accordingly.

Hint

If the sender has to stop, the throughput is the data sent divided by the time until the acknowledgment arrives.

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